MIL-S-51016A
29 MARCH 1963
SUPERSEDING
MIL-S-51016
29 JUNE 1960

MILITARY SPECIFICATION

SODIUM PHOSPHATE, MONOBASIC, ANHYDROUS BIOLOGICAL GRADE

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers a biological grade of anhydrous monobasic sodium phosphate.

2. APPLICABLE DOCUMENTS

2.1 **Government documents.** The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

RR-S-366 — Sieves, Standard for Test Procedures.

MILITARY

UU-S-48 —Sacks, Shipping, Paper.

PPP-D-723 — Drums, Fiber.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

(Copies of specification, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following document, of the issue in effect on date of invitation for bids, forms a part of this specification to the extent specified herein:

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies of these freight classification rules should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 **Material.** The material shall be a granular, biological grade, anhydrous, monobasic sodium phosphate (NaH_2PO_4). The material shall be a primary sodium phosphate and not a process byproduct.

3.2 **Chemical and physical properties.** The chemical and physical properties of the sodium phosphate shall conform to the require-

FSC 6810

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ments shown in table I when tested as specified in the test paragraphs referenced therein.

TABLE I. *Chemical and physical properties*

Properties	Requirements		Test paragraph
	Minimum	Maximum	
Phosphorus pentoxide (P_2O_5).	58.5%	59.2%	4.6.1
ph	4.4	4.8	4.6.2
Moisture	.....	0.5%	4.6.3
Clarity of 5% solution.	90%	.....	4.6.4
Arsenic (as As_2O_3)	.....	0.5 ppm	4.6.5
Heavy metals (as Pb).	.....	10.0 ppm	4.6.6
Particle size:			4.6.7
passing 2000 micron (No. 10) sieve.	98%	.....	
passing 149 micron (No. 100) sieve.	.....	30%	

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Special provisions.

4.2.1 *Alternative inspection (including testing) procedures.* The supplier may utilize any alternative inspection procedures which will insure equal or better quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.2 *Supplemental evidence.* The supplier shall provide evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 have been satisfied.

4.3 Lotting. A lot shall consist of the monobasic sodium phosphate offered for acceptance at one time, which has been produced by one manufacturer under essentially the same manufacturing conditions, provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.4 Sampling.

4.4.1 *For nondestructive examination.* Sampling shall be conducted in accordance with MIL–STD–105.

4.4.2 *For test.* Sample containers shall be taken at random from each lot as follows:

Lot size	Sample size
1	1
2– 275	2
276– 545	3
546– 900	4
901–1345	5
1346–1875	6
1876–2500	7

Sample size for lots exceeding 2500 containers shall be calculated using the following equation:

$$n = 0.15\sqrt{N}$$

where: n = Sample size.
N = Lot size.

A representative specimen shall be removed from each container in the sample and placed in a clean, dry container labeled to identify the lot and the container from which it was taken. Each specimen shall be tested as specified in 4.6.

4.5 Examination. Sample containers of monobasic sodium phosphate shall be examined for compliance with level A and B preparation for delivery requirements in accordance with the classification of defects and with MIL–STD–105.

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4.5.1 *Classification of defects.*

4.5.1.1 *Preparation for delivery (section 5).*

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Container incorrect
102	Net content of container incorrect
103	Container closure incorrect
104	Marking incorrect

4.6 **Tests.** Analytical reagent grade chemicals and distilled water shall be used throughout the tests. All weighings shall be made to four significant figures. Tests shall be conducted as follows:

4.6.1 *Phosphorus pentoxide (P_2O_5) (see 6.4).*

4.6.1.1 *Solutions.*

4.6.1.1.1 *Ammonium molybdate solution.* Dissolve 118 grams (g) of 85 percent molybdic acid in a mixture of 400 milliliters (ml) of water and 80 ml of concentrated ammonium hydroxide solution. Cool, filter if necessary, and slowly pour, with stirring, into a cooled mixture of 400 ml of concentrated nitric acid and 600 ml of water. Add about 0.05 g of dibasic sodium phosphate dissolved in a little water, mix thoroughly and allow the mixture to settle for 24 hours. Use the clear supernatant liquor, filtering if necessary. Store the solution in a cool, dark place.

4.6.1.1.2 *Methyl orange indicator.* Dissolve 0.1 g of methyl orange in water and dilute to 100 ml.

4.6.1.1.3 *Standard nitric acid (0.324 N).* Prepare a 0.324 *N* nitric acid solution using carbon dioxide-free water. Standardize against 0.324 *N* sodium hydroxide.

4.6.1.1.4 *Standard sodium hydroxide (0.324 N).* Prepare a 0.324 *N* sodium hydroxide solution using carbon dioxide-free water. Standardize against National Bureau of Standards benzoic acid. (One ml of 0.324 *N* sodium hydroxide is equivalent to approximately 0.001 g of P_2O_5.) Determine the P_2O_5 equivalent of the solution against monobasic potassium phosphate to obtain the exact value of the solution, using the general procedure specified in 4.6.1.2.

4.6.1.1.5 *Potassium nitrate solution.* Dissolve 10 g of potassium nitrate in water and dilute to 1 liter.

4.6.1.1.6 *Phenolphthalein indicator.* Dissolve 1 g of phenolphthalein in 50 ml of ethyl alcohol and add 50 ml of water.

4.6.1.2 *Procedure.* Transfer 1.4 g of the specimen into a 500-ml volumetric flask, dissolve it in water, and dilute to the mark. Thoroughly mix the solution, and pipette a 25.0 ml aliquot into a 400-ml beaker containing 75 ml of water. Add 1 drop of methyl orange indicator and neutralize the solution with dilute ammonia or nitric acid. Add 5 to 8 g of ammonium nitrate crystals, and cool to room temperature, stirring until the ammonium nitrate is completely dissolved. Add 60 ml of the ammonium molybdate solution with stirring, and then stir vigorously for a period of 3 minutes. Allow the mixture to stand for a period of 5 to 6 minutes and then filter into a 500-ml suction flask, using suction, through a diatomaceous silica filter aid coated ¼-inch paper-pulp[1] filter supported on a 1-inch perforated porcelain plate. Rinse the beaker with about 25 ml of the potassium nitrate and then pour it onto the filter. Repeat this rinsing operation five times. Finally, carefully rinse the filter five more times with the potassium nitrate solution. Transfer the precipitate and the filter pad to the beaker in which the precipitation was made and add 150 ml of water. Using a buret for measuring, add standard sodium hydroxide solution (0.324 *N*) until the yellow precipitate is dissolved and then add 5 to 8 ml more. Add 5 to 10 drops of phenolphthalein indicator and discharge the pink color by titrating with standard nitric

[1] The paper pulp suspension may be made by vigorously shaking coarse unwashed analytical filter paper in a little hot water.

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acid (0.324 *N*). Finally titrate to a perceptible pink color with the standard sodium hydroxide solution (0.324 *N*).

4.6.1.3 *Calculation*. Calculate the percent phosphorous pentoxide using the formula:

$$\text{Percent } P_2O_5 = \frac{2000\ F\ (A-B)}{W}$$

where: A = total ml of 0.324 *N* sodium hydroxide added.

B = total ml of 0.324 *N* nitric acid used.

F = equivalent value (as P_2O_5) of 0.324 *N* solution.

W = weight of specimen used in g.

4.6.2 *pH*. Dissolve 1.0 g of the specimen in 100 ml of recently boiled water. Determine the pH of the solution at 20° ± 1°C., using a glass electrode-saturated calomel electrode-pH meter system (see 6.5) in accordance with the instruction on the meter.

4.6.3 *Moisture*.

4.6.3.1 *Procedure*. Weigh 2 g of the specimen into a tared aluminum or glass weighing dish equipped with a cover. Place the dish, with the cover removed, in an oven at a temperature of 105° ± 2° C. for a period of 2 hours. At the end of the drying period, place the uncovered dish in a desiccator to cool, replace the cover, and reweigh.

4.6.3.2 *Calculation*. Calculate the percent moisture using the formula:

$$\text{Percent moisture} = \frac{100\ (W-Y)}{W}$$

where: W = initial weight of specimen.

Y = dry weight of specimen.

4.6.4 *Clarity of 5 percent solution*. Standardize a photoelectric colorimeter at 100 percent light transmission against water in a 50 millimeter (mm) optic cell, using a red filter having a wave length of approximately 650 millimicrons. Dissolve 5 g of the specimen in 100 ml of water, and allow the solution to stand for 10 minutes with occasional swirling. At the end of the 10-minute period, transfer the solution to a similar 50-mm optic cell and place it in the colorimeter. Determine the percent light transmission by reading the meter.

4.6.5 *Arsenic (as As_2O_3) (Gutzeit method)*.

4.6.5.1 *Reagents and materials*.

4.6.5.1.1 *Lead acetate solution*. Dissolve 10 g of lead acetate in 100 ml of water. (If the solution is cloudy, add the minimum quantity of acetic acid required to clear the solution.)

4.6.5.1.2 *Stannous chloride solution*. Dissolve 40 g of arsenic-free stannous chloride ($SnCl_2.2H_2O$) in concentrated hydrochloric acid and dilute to 100 ml with concentrated hydrochloric acid.

4.6.5.1.3 *Potassium iodide solution*. Dissolve 15 g of potassium iodide in water and dilute to 100 ml with water.

4.6.5.1.4 *Sand*. Use clean 30-mesh (through No. 30 and retained on No. 40) white sea sand. Clean the sand by washing successively with hot 10 percent sodium hydroxide solution, hot concentrated nitric acid, and hot water. Dry the washed sand.

4.6.5.1.5 *Mercuric bromide paper*. Use commercial arsenic test papers of uniform weight and texture cut into strips 2.50 mm wide and about 120 mm long. Sensitize strips by soaking them in a 5-percent filtered solution of mercuric bromide in 95 percent ethyl alcohol, for a period of 1 hour. After sensitization, nearly dry the strips on glass rods. (Handle the strips with forceps at all times after sensitization.) While the strips are nearly dry, place them between sheets of clean paper and subject them to pressure long enough to eliminate bends and curls. Store excess strips in a dry, dark place for a period of not more than 90 days.

4.6.5.1.6 *Standard arsenic solution*. Dissolve 1.00 g of arsenic trioxide in 25 ml of 20 percent sodium hydroxide solution. After com-

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plete solution, saturate the solution with carbon dioxide and dilute to 1,000.0 ml with water. (1 ml = 1 milligram (mg) As_2O_3). Dilute 40.0 ml of this solution to 1,000.0 ml with water. Pipet 50.0 ml of the latter dilution into a 1-liter volumetric flask and dilute to 1,000.0 ml. (This dilution contains 0.002 mg of As_2O_3 per ml.)

4.6.5.1.7 *Zinc.* Use 20- or 30-mesh arsenic-free granulated zinc, or arsenic-free stick zinc either cut into equal 10-mm lengths or melted and cast into pellets in a porcelain mold approximately 9 mm in diameter and 12.5 mm deep. Activate the cut or cast pellets of zinc with dilute hydrochloric acid (1:3) to which has been added 2 ml of the stannous chloride solution, allowing the action to continue for a period of 15 minutes. Sort out inactive or overactive pieces. Wash the remaining pellets free from acid with water. Select uniformly etched nonpitted pellets and store them in a suitable receptacle.

4.6.5.2 *Apparatus.*

4.6.5.2.1 *Generator.* Select 2-ounce wide-mouth bottles of uniform capacity and design. Mark the bottles at the 40-ml volume and fit each with a tight-fitting one-hole rubber stopper for connection with the scrubbing tube.

4.6.5.2.2 *Scrubbing tube.* Prepare a glass tube, 1 centimeter (cm) in diameter and 6 to 7 cm long, constricted at the bottom end to facilitate connection with the generator. Place a small wad of glass wool in the constructed bottom of the tube and add 3.5 to 4.0 g of the sand, taking care to have the same quantity in each tube. Moisten the sand in each tube with the lead acetate solution and remove any excess by light suction. (If the sand becomes dry or dirty, clean the sand, in the tube, by washing with concentrated nitric acid, washing with water, and suction followed by treatment with lead acetate solution.)

4.6.5.2.3 *Reaction tube.* Connect a glass tube 2.6 to 2.7 mm in diameter and 100 to 120 mm long to the top end of the scrubbing tube by means of a tight-fitting one-hole rubber stopper. Clean and dry the reaction tube (an ordinary pipe cleaner may be used) before inserting the strip of the mercuric bromide paper.

4.6.5.3 *Procedure.* Place 1 g of the specimen in a generator containing 10 ml of water. Make the solution acid to brom cresol green with concentrated hydrochloric acid, then add an excess of 5 ml of concentrated hydrochloric acid, and cool to room temperature. Add 5 ml of the potassium iodide solution and 4 drops of the stannous chloride solution. Prepare a standard by pipetting 1.0 ml of the standard arsenic solution into a separate generator and adding 5 ml of concentrated hydrochloric acid, 5 ml of the potassium iodide solution, and 4 drops of the stannous chloride solution. Mix the solutions, allow them to stand for 30 minutes at a temperature of 20° to 25° C., and dilute them to 40 ml with water. Handling the mercuric bromide paper with forceps, cut the strips in half and insert one-half strip (cut surface down) into each reaction tube, carefully centering each strip in each tube. Add 2 to 5 g of granulated 20 to 30-mesh zinc, or 10 to 15 g (two pellets) of activated stick zinc (cut lengths or pellets), depending on the activity of the zinc. (Add the same quantity to each generator.) Immediately assemble each apparatus, immerse to within 1 inch of the top of the reaction tube in a water bath at a temperature of 20° to 25° C. Allow the evolution of gas to proceed for a period of 90 minutes. Remove the strips and compare the stains caused by the specimens with that caused by the standard. (If the stain caused by the specimen is a deeper color or longer than that caused by the standard, the sample represented by the specimen does not comply with the requirement specified in 3.2).

4.6.6 *Heavy metals (as Pb).*

4.6.6.1 *Reagents.*

4.6.6.1.1 *1 N Acetic acid.* Dilute 60 ml of glacial acetic acid to 1,000 ml with water.

4.6.6.1.2 *Standard lead solution.* Dissolve 0.1598 g of lead nitrate in 100 ml of water containing 1 ml of concentrated nitric acid and

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dilute the solution to 1,000.0 ml with water. Dilute 10.0 ml of this solution to 100.0 ml. (1 ml contains the equivalent of 0.01 milligrams of lead.)

4.6.6.2 *Procedure.* Dissolve a 4.000 g of the specimen in 25 ml of water, add 2 ml of 1 *N* acetic acid and dilute to 40 ml. Transfer 10.0 ml of this solution into a matched comparison tube, add 2.0 ml of the standard lead solution and dilute to 40 ml (control solution). Dilute the remaining 30 ml of the solution of the sample to 40 ml in a second matched comparison tube (test solution). Add 10 ml of freshly prepared saturated hydrogen sulfide solution in water to each matched comparison tube, mix the contents of the tubes, and compare the colors of the solutions in a comparator. (If the test solution is darker than the control solution, the sample represented by the specimen does not comply with the requirement specified in 3.2.)

4.6.7 *Particle size.*

4.6.7.1 *Procedure.* Assemble a 2000-micron (No. 10) and a 149-micron (No. 100) sieve, both conforming to Specification RR-S-366, with the 2000-micron sieve on top, and fit them with a cover and receiving pan. Remove the cover, place 100.0 g of the specimen on the 2000-micron sieve, and replace the cover. Shake the nest of sieves for 5 minutes by means of a single eccentric type, mechanical sieve shaker at a uniform speed of 300 ± 15 gyrations and 150 ± 10 taps per minute. Weigh the residues on the 149-micron sieve and in the pan.

4.6.7.2 *Calculation.*

$$\text{Percent passing 2000 micron (No. 10) sieve} = \frac{100\ (A + B)}{W}$$

$$\text{Percent passing 149 micron (No. 100) sieve} = \frac{100\ B}{W}$$

where: A = weight of residue on 149 micron sieve.

B = weight of residue in pan.

W = weight of specimen taken for sieve test.

4.7 **Rejection criteria.** If any specimen of monobasic sodium phosphate fails to meet the test requirements of this specification, the lot represented shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 *Level A.* One hundred pounds of monobasic sodium phosphate shall be packed into fiber drums or paper sacks. The drums shall conform to type III, grade A, of Specification PPP-D-723, and shall be furnished with a water vapor barrier or liner for protection against deterioration of the contents. The water vapor barrier or liner shall not adversely affect or be adversely affected by the contents of the drums. If a metallic vapor barrier or liner is used, it shall not be in direct contact with the contents. Sacks shall conform to construction number 14X (with dipped ends) of Specification UU-S-48. Any water vapor barrier other than asphalt may be used provided that it equals or exceeds all of the requirements of table V — class C, asphalt laminated shipping sack kraft paper, of Specification UU-S-48.

5.1.2 *Level B.* Sodium phosphate shall be packaged as specified in 5.1.1 except that the fiber drums shall conform to type I, grade A, of Specification PPP-D-723 and the sacks shall conform to construction number 10 *L/W* (with dipped ends) of Specification UU-S-48.

5.1.3 *Level C.* One hundred pounds of monobasic sodium phosphate shall be packed in metal containers, fiber drums or multiwall paper bags in a manner to insure carrier acceptance and safe delivery at destination. Metal containers shall be fitted with liners. Fiber drums or multiwall paper bags shall have a barrier, liner, or interior coating, as applicable, to afford protection against water vapor. Interior liners and coatings shall be nonmetallic, and shall be composed of materials which will not adversely affect or be adversely affected by the contents. Containers shall be in accordance with Uniform Freight Classification Rules or regulations of other carriers ap-

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plicable to the mode of transportation. Containers supplied under the same contract shall be of uniform size, shape, and material.

5.2 Marking. In addition to any marking required by the contract or order, all shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. This specification covers anhydrous monobasic sodium phosphate which is intended for use in biological media. The product is similar to, but does not include all of the requirements established for a food grade of monobasic sodium phosphate.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Level of packing required.

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing process intended to make the final product substantially uniform.

6.4 Phosphorus pentoxide. The test method specified in 4.6.1 is considered to be adequate for routine determinations of phosphorous pentoxide. In disputed cases, the method for the determination of phosphorous pentoxide specified in ASTM Designation D 501 should be used.

6.5 pH meters and systems. pH meters and systems complying with the requirements of ASTM Designation E70 have been found to be satisfactory for pH determination.

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